



USER MANUAL

PROJECT TITLE:

**DESIGN AND IMPLEMENTATION OF A DRIVER
DROWSINESS DETECTION AND SPEED CONTROL SYSTEM
FOR TRUCK DRIVERS**

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1.0 Introduction

This project is a real-time Driver Drowsiness Detection and Speed Control System designed to help reduce accidents caused by driver fatigue. It uses computer vision and physiological signals to detect drowsiness in truck drivers. Upon detection, the system issues visual and audio alerts and slows down the vehicle’s throttle using a simulated motor control system.The core components include:

- Raspberry Pi 4
- USB Webcam
- GY-MAX30102 Pulse Oximeter Sensor
- 1602 I2C LCD Display
- Buzzer and LEDs for alerts
- DC Geared Motor with MOSFET-based speed control
- 12V Power Supply with Buck Converter

2.0 System Overview and Features

Feature	Description
Drowsiness Detection	Uses Eye Aspect Ratio (EAR) and PERCLOS from camera; heart rate from MAX30102
LCD Display	Shows heart rate and status messages like “Drowsiness Detected”
Audio & Visual Alerts	Buzzer and LEDs signal the driver when drowsiness is detected
Motor Control	Simulates throttle reduction using PWM to slow down a DC motor
Real-Time Monitoring	Constant face detection and heart rate reading
Two-Stage Intervention	First alerts the driver, then slows motor if drowsiness persists
Safety Mechanisms	Reverse polarity diode, voltage regulation, and signal filtering

3.0 Operation Instructions

3.1 System Startup

1. Connect 12V battery input to the system.
2. Ensure the Raspberry Pi, camera, LCD, and MAX30102 sensor are connected properly.

3. Power ON the system.
4. The system initializes and starts reading camera and heart rate data.
5. Green LED turns ON to indicate normal operation.
6. LCD displays “Monitoring...” and heart rate in BPM.

3.2 Drowsiness Monitoring

- The camera captures facial landmarks using OpenCV and Dlib.
- EAR (Eye Aspect Ratio) and heart rate are continuously evaluated.
- If $EAR < \text{threshold}$ and/or heart rate is low:

Buzzer sounds

Red LED turns ON

LCD displays “Drowsiness Detected”

3.3 Speed Control Intervention

If drowsiness persists for a few seconds without user response:

- The system stops or reduces the speed of the DC motor using PWM.
- LCD displays “Motor Slowing Down.”
- Green LED turns OFF

3.4 Buzzer Alert System

- When battery percentage drops **below 20%**, Voltage display turns RED.
- The system will continue to update battery data during this time.

4.0 Maintenance Tips

- Keep the camera lens clean for accurate face detection.
- Ensure good finger placement on the MAX30102 sensor.
- Periodically check wire connections and terminals.
- Do not operate system in extreme heat, dust, or moisture.
- Ensure proper ventilation around Raspberry Pi.

5.0 Troubleshooting Guide

Issue	Possible Cause	Solution
LCD not displaying	Raspberry Pi not powered	Check 5V supply and buck converter wiring
Buzzer doesn't sound	GPIO or transistor miswiring	Check buzzer and driver circuit
Motor not responding	PWM not being sent / MOSFET not switching	Check GPIO pin, MOSFET gate driver
No camera feed	USB camera not detected	Reconnect or try different USB port
MAX30102 not reading	Loose I2C connection	Recheck SDA/SCL wires and sensor placement

6.0 Technical Summary

Parameter	Value
Processor	Raspberry Pi 4 (Quad-core ARM Cortex-A72)
Camera	USB Webcam (30fps, 720p)
Drowsiness Indicators	EAR, PERCLOS, heart rate (MAX30102)
Display	1602 I2C LCD (5V, 2-wire interface)
Alert System	Buzzer + Red/Green LEDs
Speed Control	12V DC motor controlled via IRF4905 MOSFET
Power Supply	12V DC input, LM2596 Buck Converter to 5v
Software	Python with OpenCV, Dlib, and RPi.GPIO

7.0 Conclusion

This project successfully demonstrates a real-time embedded system capable of detecting driver drowsiness using facial and physiological data. The integration of visual/audio alerts and motor control makes it a strong prototype for future implementation in trucks or commercial vehicles. The use of low-cost hardware and open-source libraries ensures replicability and adaptability in similar safety-critical applications.